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INTRODUCTION TO RESEARCH DATA MANAGEMENT

Materials in chemistry and physics 2023-2024, Paula Oset, 1st December 2023
https://osf.io/jcf42

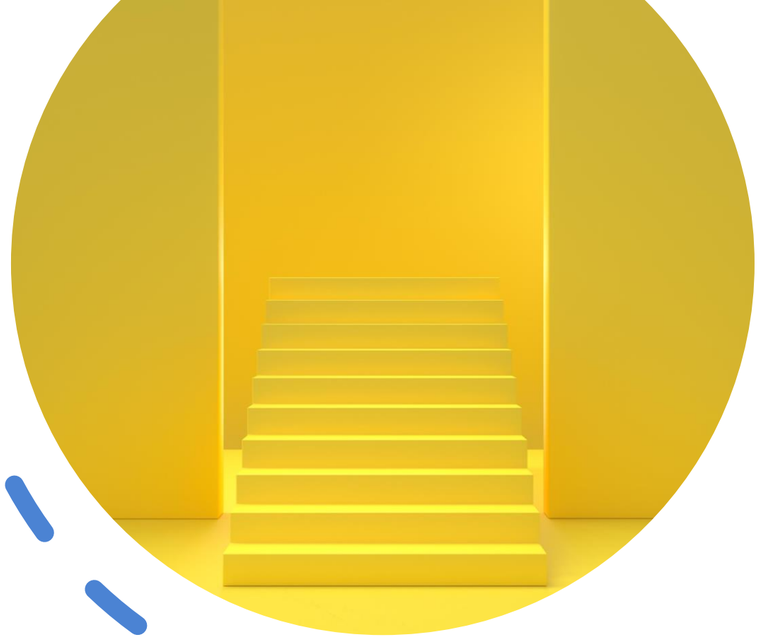
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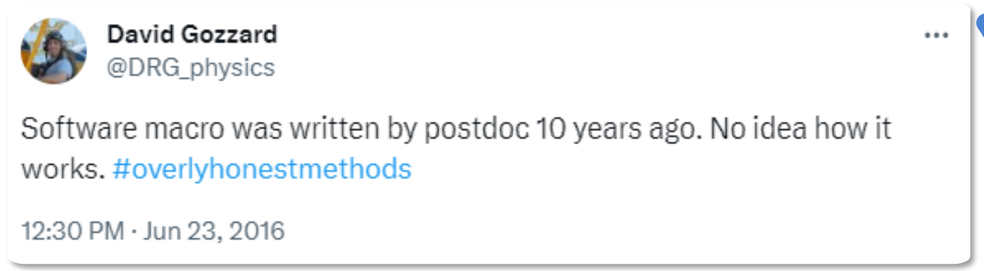
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OUTLINE

- Why RDM?
- The data lifecycle
  - Best practices
  - Tips & tools
- Planning, storing, preserving & sharing



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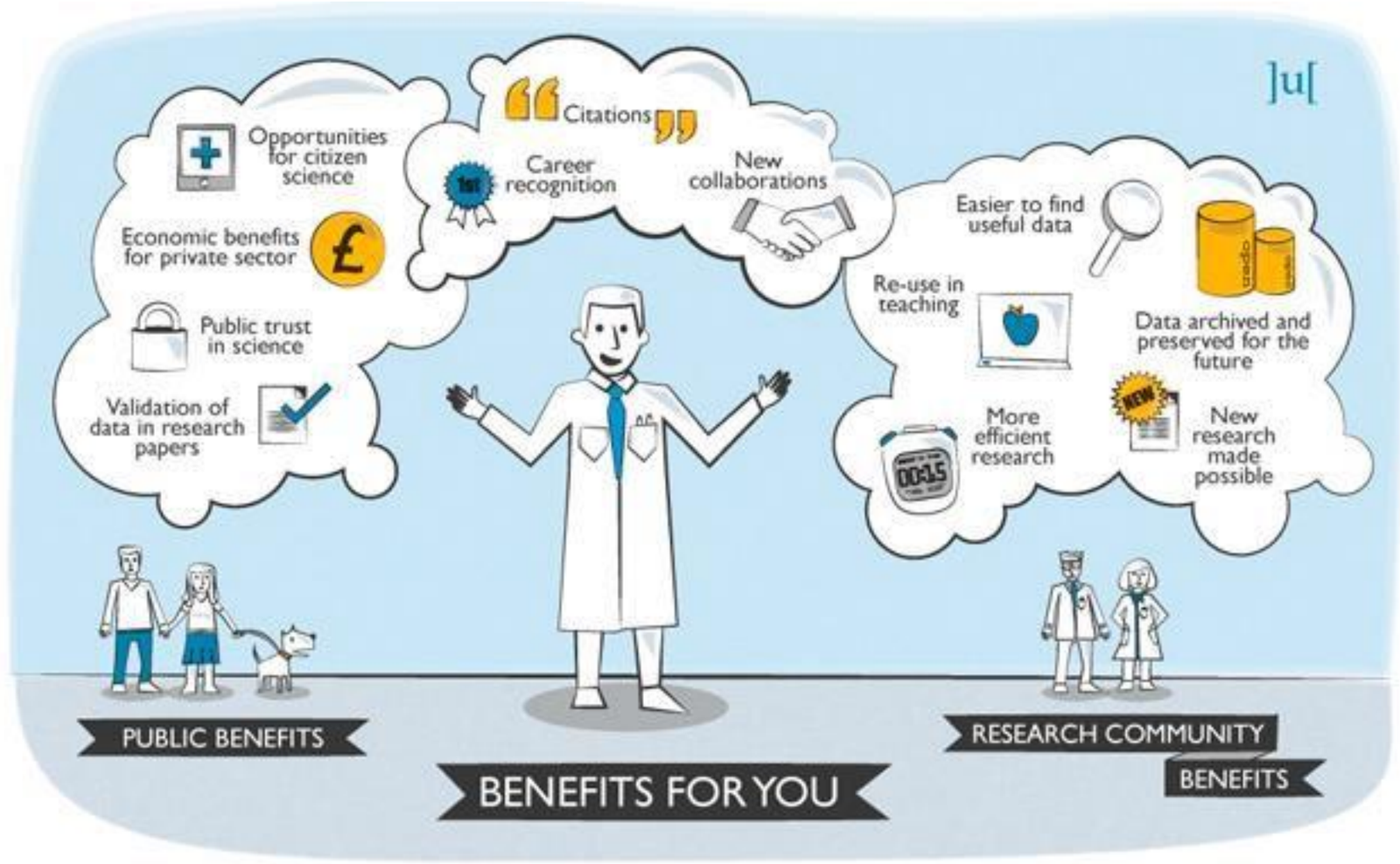


WHAT COULD GO WRONG?

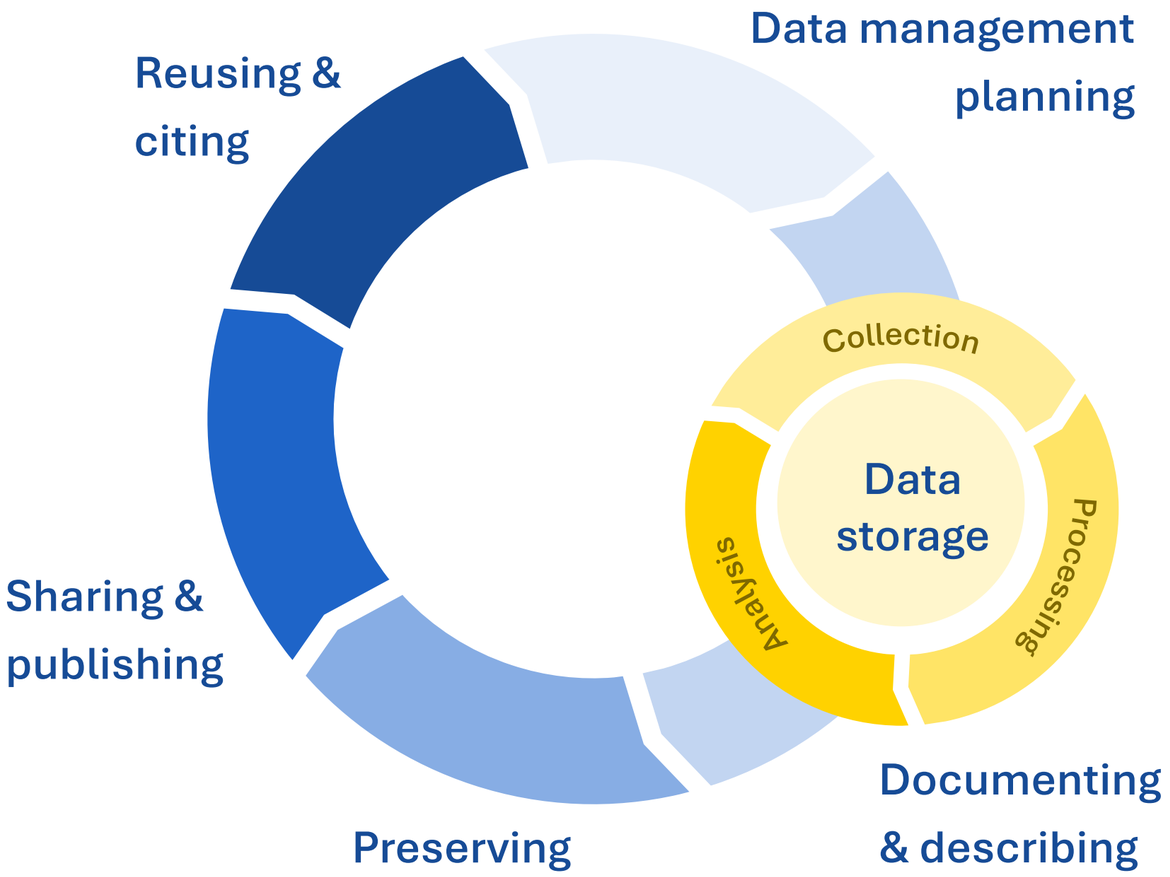
- 3 years into the project, data is lost
- Data exists but can't be reused (no documentation, corrupted, outdated format…)
- You are required to share (personal) data but you didn't ask for consent
- Disagreement between project partners on exploitation of results
- Can't keep track of your own files

MANAGING AND SHARING DATA: THE BIGGER PICTURE

RDM: Taking care of data so that it remains (effectively) reusable



RESEARCH DATA LIFECYCLE











Adapted from DCC

Data management planning



DATA MANAGEMENT PLAN (DMP)

- Outlines how research data will be handled **during** & **after** a project.
- Increasingly required by research funders/institutions

  -> as a **first step** towards good research data management (RDM)

  -> requires times up-front, saves time & problems in the long run
- DMPonline.be
- DMP examples
  - By UGent researchers, curated by UGent data stewards



DMP SECTIONS



Data summary & description

Purpose, origin, type, sensitiveness, format & volume



Documentation & metadata

Information to keep data findable, understandable, reusable



Roles & responsibilities



Ethical & legal issues

Intellectual property, GDPR, etc.



Long-term preservation

What to preserve & where, for how long, access restrictions



Costs & resources



Data storage & security

Security measures, back-up, data transfers & access control



Data sharing and reuse

Data repositories, access conditions, licenses or agreements



Storing data safely

THE DO'S AND DON'TS OF DATA SECURITY

Avoid local storage or own IT resources

Avoid storing confidential information in third-party cloud services

Do not send data by email

Use institutional storage

Backup regularly

Configure access permissions

Encrypt confidential data

Use safe file transfer alternatives

Keep your system updated

Use strong passwords

Protect your hardware

Control physical access to confidential data

When needed, destroy information in a safe way

- Shred paper
- Erasing software
- Destroy hardware

More: https://www.ugent.be/en/research/datamanagement/during-research/data-security.htm

Knowledge clip: Data security

RECOMMENDED STORAGE OPTIONS OFFERED BY UGENT



- **SharePoint sites:** cloud based storage offered by UGent DICT
- Large **capacity** but individual file size <250GB
- Easy to **synchronize** to local: accessible anywhere via browser or from file explorer (also offline)
- Share with external users
- Give access to (at least) your promotor
  - (vs. OneDrive) Avoid data is lost when/if your account is suspended
- Automatic **back-up and version control**
- **RDM workspace template** (How to request)
- More info & demo recording



- **Shares:** network storage offered by UGent DICT. Server in S10
- Also known as *vakgroep* shares, project shares, ACL shares, UGent central disk space…
- Large capacity, no file size limit
- Need to be connected to UGent network: work on campus or with VPN
- No offline work
- Automatically backed-up: easy to recover files
- How to request
- More info & demo recording

WHAT IF…?

- **Local storage is inevitable** (pc, laptop, USB, external hardrive…) : field work, device or instrumentation limitations…
  - Higher risk of data loss
    - Make back-ups and make them happen automatically: sync software (e.g. FreeFileSync)
  - Higher risk for confidential data
    - Account password does not protect data/files on laptop's memory disk
    - Encrypt!
- You work with very large data -> is HPC an option?
- You work with your department solution: get informed! (Quota? Back-ups? Guidelines?)
- You need more than one storage platform:
  - Avoid when possible
  - Make clear arrangements: what goes where

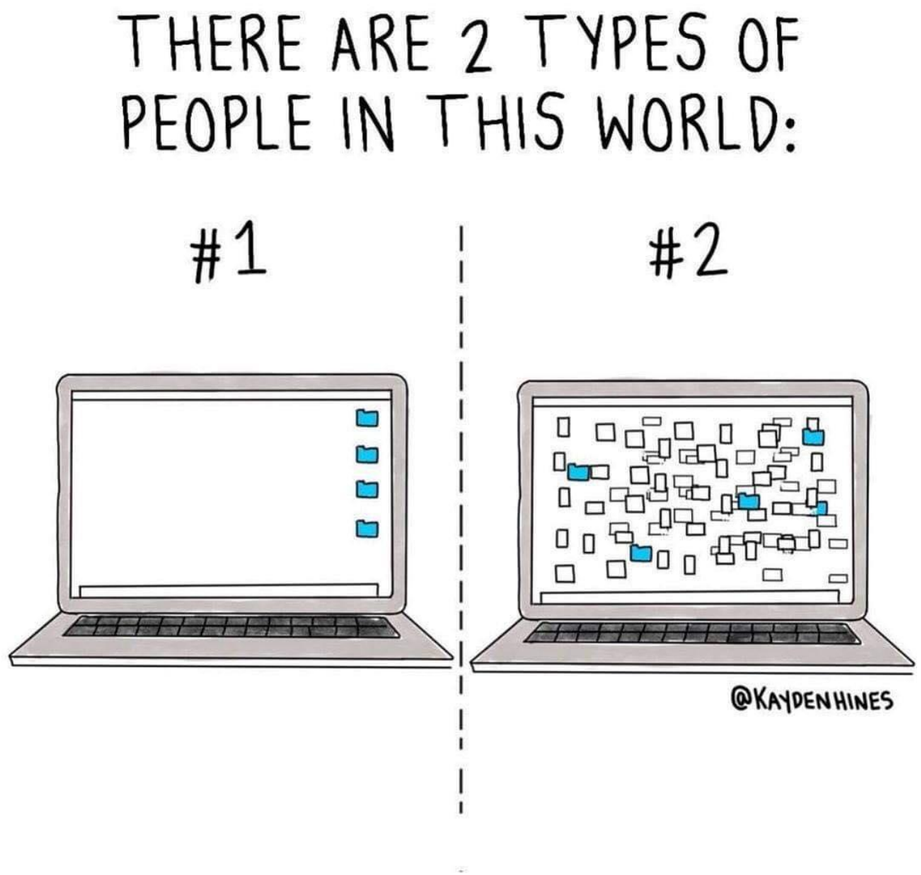
POSSIBLE WORKFLOWS

Sensitiveness of data & security measures

Main storage location	Back-up	Access control	Encryption
SharePoint site	Auto Sync to local As extra layer	Configurable (permission levels)	Encrypt local copies
Shares (network drives)	Auto As extra security layer	Configurable (ACL lists)	As extra security layer
Local storage	Manual Sync to safe location (shares) As extra security layer	Not supported	Encrypt local copies

FILE ORGANIZATION: HOW

- Make a file organization plan
  - Develop file naming convention
  - Plan folder hierarchy (and/or metadata scheme –SharePoint)
  - Develop version control strategy
- Document the file organization system
- Use the system consistently



Adapted from Harvard RDM working group, CC-BY (https://osf.io/hjntb/)

DEVELOP FILE NAMING CONVENTION

- Think of the current and expected files
  - Use unique characteristics reflecting the file content, possibly status and version
  - File name = principal identifier of file, used to uniquely identify a file
  - Be consistent

Data type	Unique characteristics	Multiple files or versions?	File format	Draft naming schema
Microscope image	Date collected Experiment number Sample type Microscope settings	>50 images per day	TIFF	YYYYMMDD_exp01_samplecode_001 YYYYMMDD_samplecode_exp01_001



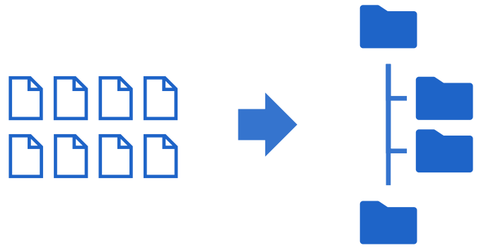
DMP: data summary or data description

Adapted from *Worksheet: Naming and Organizing your Files and Folders by MIT Libraries Data Management Services*, CC-BY,

FILE NAMING: BEST PRACTICES

LENGTH	Avoid long names, remove unnecessary elements			
SPECIAL CHARACTERS	☒	Spaces Special characters: \ / : * ? " < > | ! % & - ; = () + , .	☒	name date v1.txt name.date.v1.txt name/date&v1.txt
	☑	Underscores camelCase	☑	name_date.txt nameDate.txt
DATES	☑	Use ISO8601 YYYYMMDD	☑	20201008_name_v01.txt
NUMBERING	☑	Use leading zeroes for sequential numbering	☒ ☑	nameDate_1 nameData_001.txt, nameDate_010.txt

PLAN FOLDER HIERARCHY



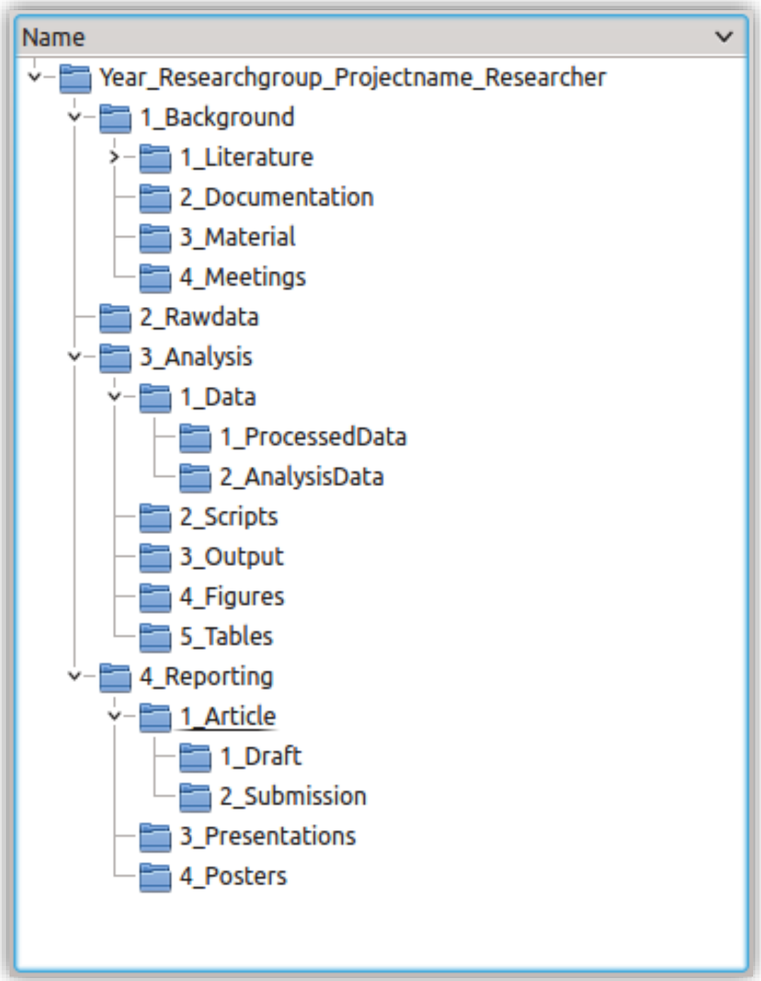
File inventory Folder hierarchy

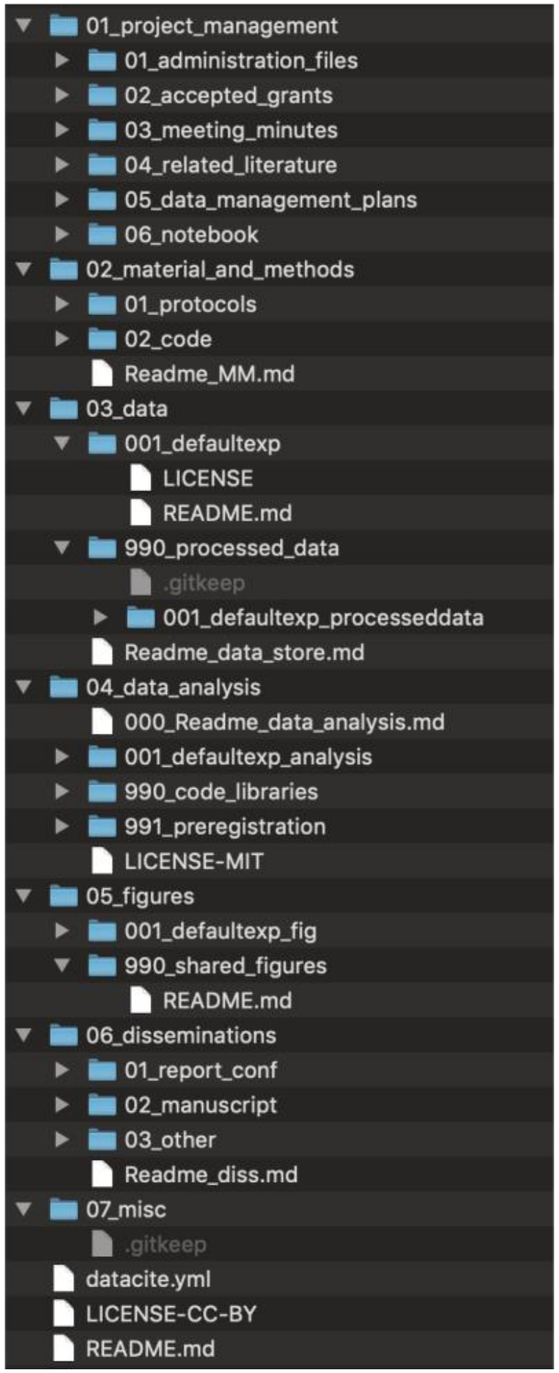
- Identify ways to divide files into categories or attributes
  - Project: one project = one folder
    - Date / time
    - Location
    - Instrument or method
    - Project stage
    - Data type: data (raw vs. processed), processing code, results, findings
- What is the most important attribute? -> Top level organization
- Create nested folders according to importance / usefulness of category
- Avoid overlapping categories and keep file names in mind
- Balance: folders not too large vs. structure not too deep

Adapted from Harvard RDM working group, CC-BY (https://osf.io/hjntb/)

PLAN FOLDER HIERARCHY: EXAMPLES

- README.txt (project context)
- Literature
- Data
  - README.txt (data structure)
  - 01_raw
  - 02_processed
  - 03_analysed (figures and tables)
  - 04_external
- Reports
  - 01_draft
  - 02_submission
- Meetings
- Scripts

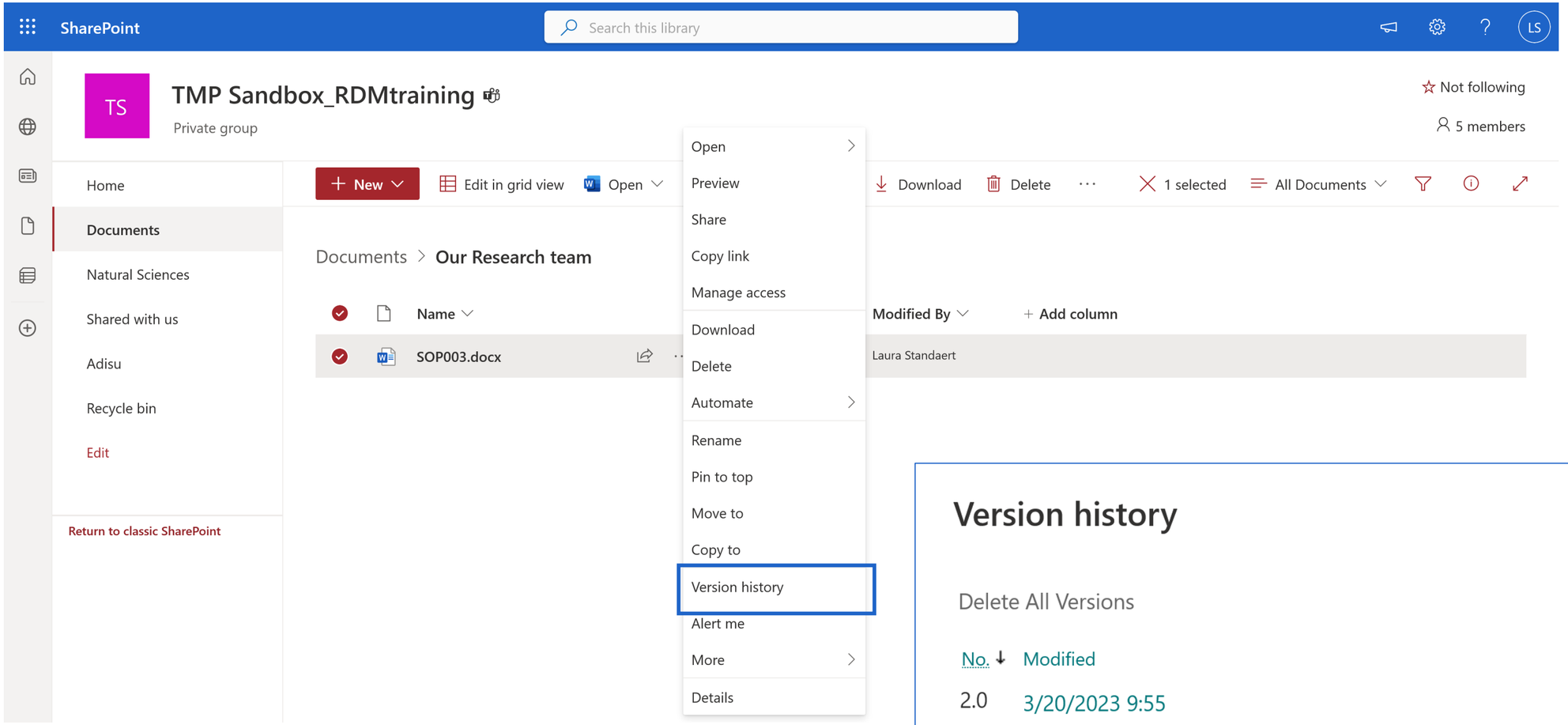


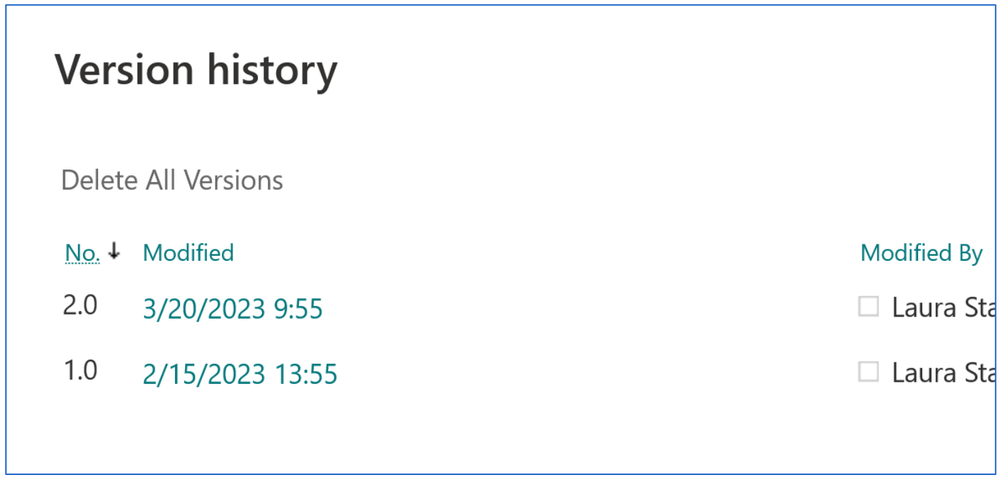


VERSION CONTROL – BEST PRACTICES

- When possible, use platforms with **automatic change logs (SharePoint, GitHub/Git…)**
- If not,
  - identify **milestone versions** of files to keep.
  - avoid clutter
  - use a systematic naming convention
  - record version and status (e.g. raw, cleaned) of a data file
- When collaborating
  - use a common ‘workspace’ (e.g. shared folder) to avoid different versions of files on different locations
  - do not use e-mail to share data

VERSION HISTORY IN SHAREPOINT





Documenting and describing





WHY DOCUMENTATION?

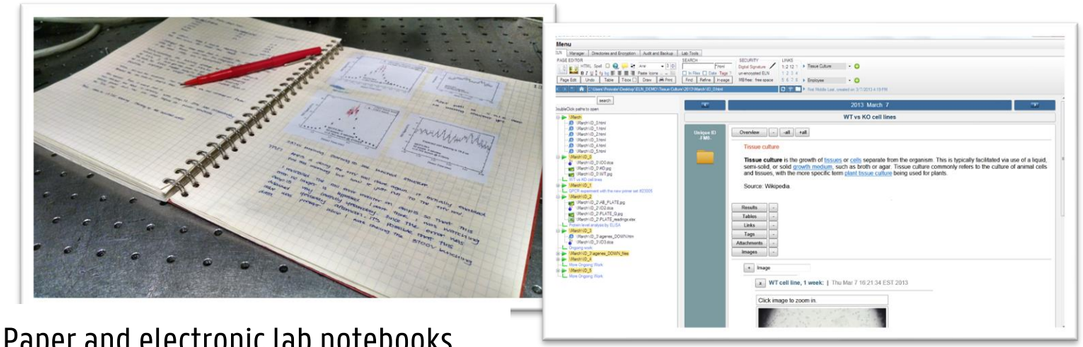
- Help yourself & others:
  - “Your future self will be your first collaborator”
  - Will you remember or will others understand…
    - what your variables mean?
      - E.g. "sglmemgp" = single member of group
    - the procedures used to transform or derive variables?
  - Keep data understandable & reusable
- Help science:
  - Research review & verification
  - The reproducibility crisis

LEVELS OF DATA DOCUMENTATION

DOCUMENTATION LEVEL	WHAT TO DOCUMENT
Item Variable	Methods, units, possible values, synonyms, abbreviations
Sample Study object	Study object, method, instrument settings, time, place
Project or study	Subject, grant, contributors, methods, process, workflow Files & relations

Variable	Definition
YEAR_WEEK	Week (Monday to Sunday) including all vaccines that were administered by the end of that week and registered in Vaccinnet+
NIS5	NIS-code (municipality) of residence of people receiving the vaccine (5 digits)
AGEGROUP	Age group of people receiving the vaccine
DOSE	Dose identifier (for vaccines requiring 2 doses: A for first dose, B for second dose; C for vaccine requiring only 1 dose; E for extra/booster dose of vaccine administered since week 36 of 2021)
CUMUL	Cumulative number of peop[illegible] than 10 vaccinations are ind[illegible] are not shown)

Sciensano COVID data codebook



Paper and electronic lab notebooks

```
project_name/
├── README.md              # overview of the project
├── data/                  # data files used in the project
│   ├── README.md          # describes where data came from
│   └── sub-folder/        # may contain subdirectories
├── processed_data/        # intermediate files from the analys
├── manuscript/            # manuscript describing the results
├── results/               # results of the analysis (data, tab
├── src/                   # contains all code in the project
│   ├── LICENSE            # license for your code
│   ├── requirements.txt   # software requirements and dependen
│   └── ...
└── doc/                   # documentation for your project
    ├── index.rst
```

Readme files

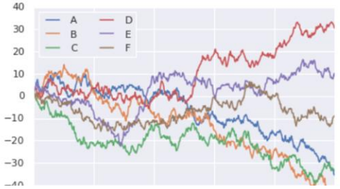
```
[5]: import matplotlib.pyplot as plt
     plt.style.use('classic')
     %matplotlib inline
     import numpy as np
     import pandas as pd
     import seaborn as sns
     sns.set()

[6]: rng = np.random.RandomState(0)
     x = np.linspace(0, 10, 500)
     y = np.cumsum(rng.randn(500, 6), 0)
```

Next step

Now, create a graph.

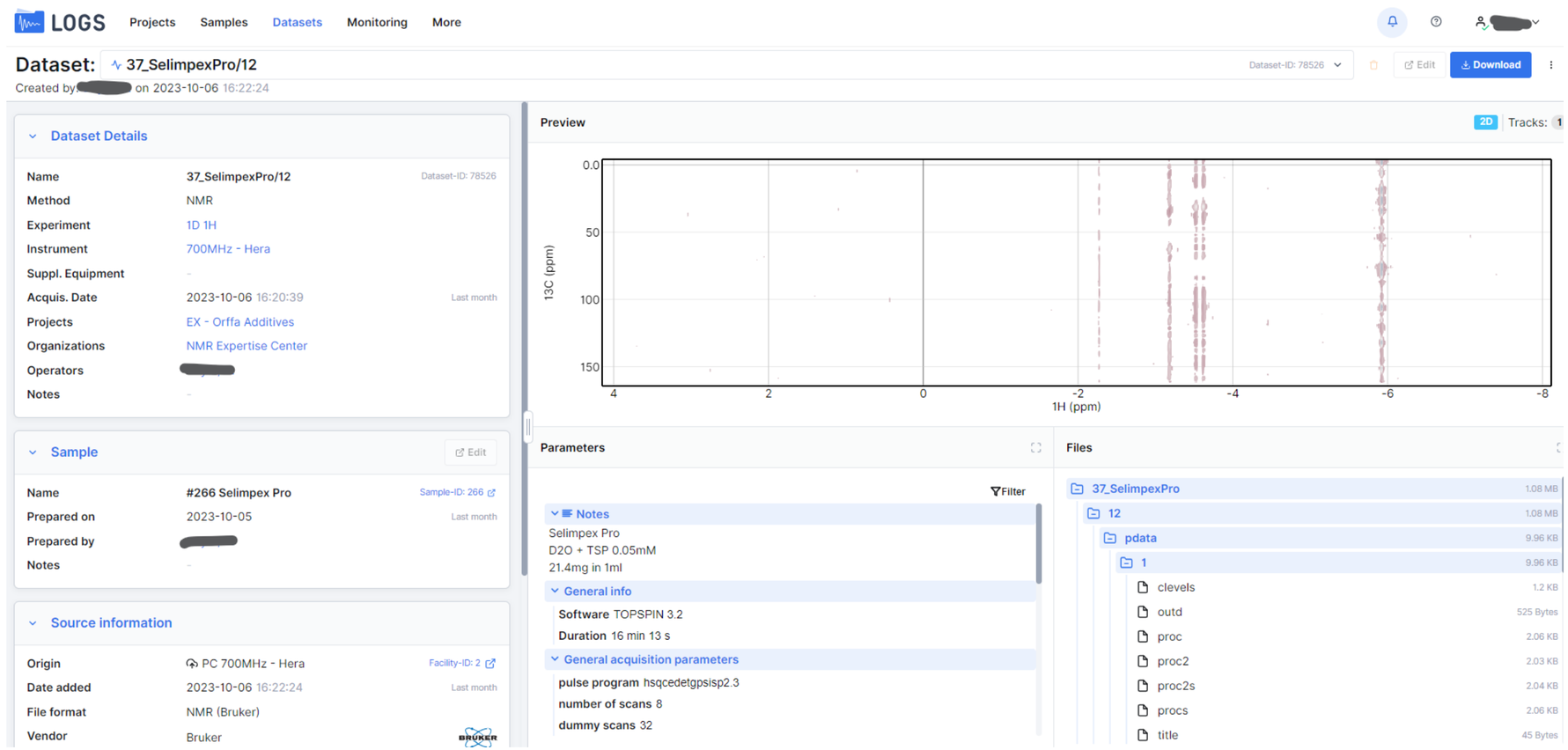
```
[7]: plt.plot(x, y)
     plt.legend('ABCDEF', ncol=2, loc='upper left');
```



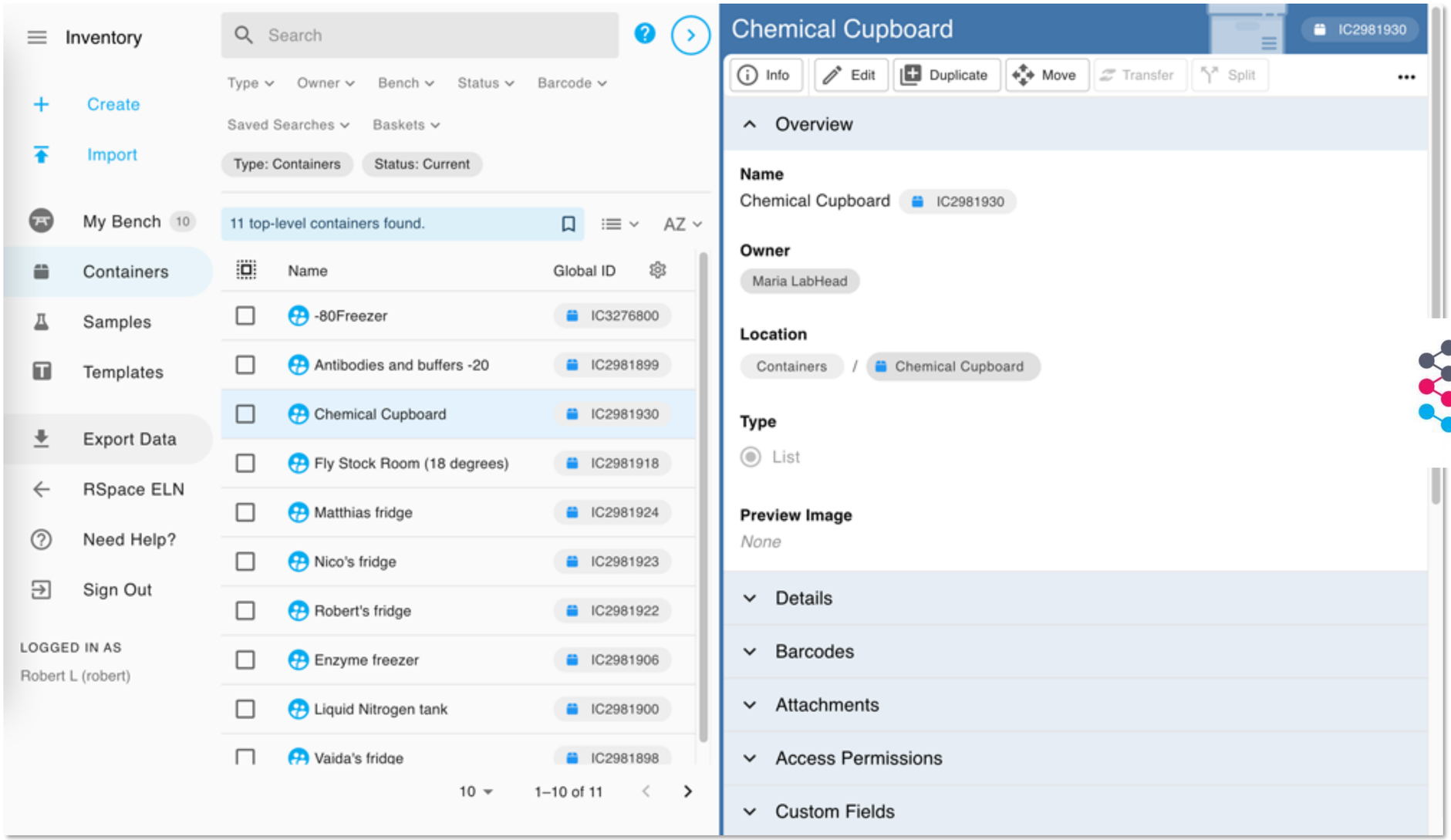
Code annotations & notebooks (e.g. Jupyter)

EXAMPLE: LOGS

- Automatically uploads raw data from instruments, alongside with metadata, and makes it accessible via your browser
- NMRS, HPLC, GC, UV-vis, IR
- Paid license
- Used by NMR Expertise Centre



E-LAB NOTEBOOK EXAMPLE: R SPACE

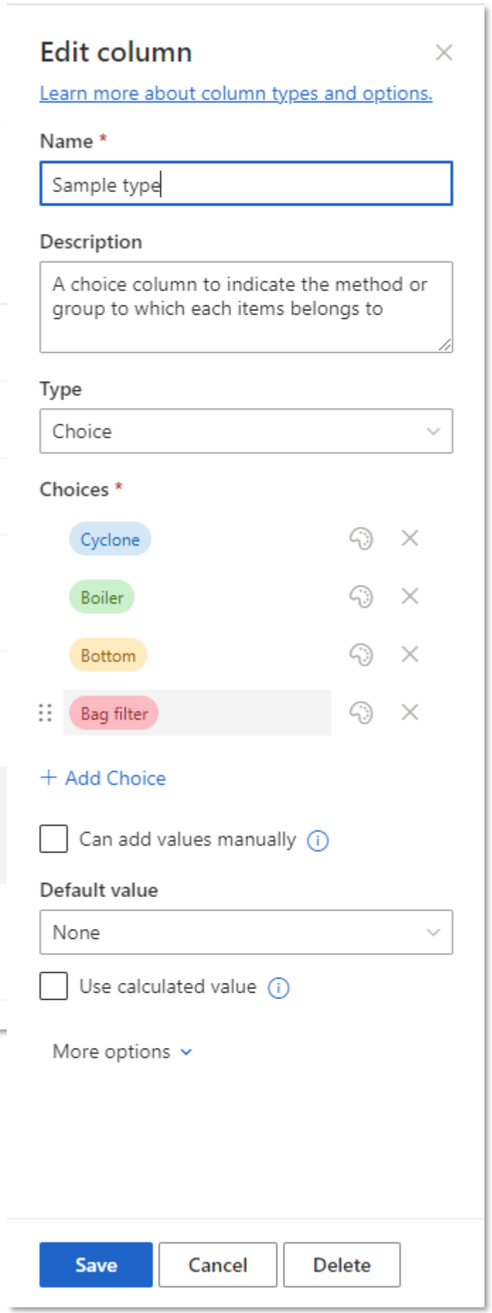


RDM WORKSPACE (SHAREPOINT TEMPLATE)

Sample inventory ☆

Title	Sample ID	Date	Sample type	Sample origin	Treatment
Sample1 Cyclone ash single	Sample 1	19/10/2022	Cyclone	sewage sludge	mono-combustion
Sample 2 Boiler ash single	Sample 2	20/10/2022	Boiler	sewage sludge	mono-combustion
Sample 3 Bottom ash single	Sample 3	21/10/2022	Bottom	sewage sludge	mono-combustion
Sample 4 Boiler ash combined	Sample 4	20/10/2022	Boiler	sewage sludge municipal solid waste	co-combustion
Sample 5 Cyclone ash combined	Sample 5	19/10/2022	Cyclone	sewage sludge municipal solid waste	co-combustion
Sample 6 Bag filter	Sample 6	21/10/2022	Bag filter	sewage sludge municipal solid waste	co-combustion
Sample 7 Bottom ash combined	Sample 7	21/10/2022	Bottom	sewage sludge municipal solid waste	co-combustion

Example: [Nutrient Recycle RDM workspace]



OSF FOR OPEN RESEARCH & DOCUMENTATION

OpenLPE | Metadata | Files | Wiki | Analytics | Registrations

OpenLPE

Contributors: Jason Stafford, Jacob Brown, Diego Perez-Alvarez

Date created: 2023-02-22 04:50 PM | Last Updated: 2023-09-14 01:13 PM

Identifier: DOI 10.17605/OSF.IO/EVUPY

Category: Project

Description: *Open-source repository of information on the manufacturing of two-dimensional materials (2DMs) using a low-cost, scalable, high shear mixing approach.*

License: *GNU General Public License (GPL) 3.0*

Wiki

OpenLPE

OpenLPE is an open-source repository of information on the synthesis of 2D materials via Liquid Phase Exfoliation (LPE) using a low-cost, shear-assisted method.

Purpose

We aim to democratise access to atomically thin materials for researchers everywhere, in a cost-effective way. To achieve this we have:

- Created open-source labware for materials synthesis
- Demonstrate detailed characteris...

Read More

Files

Citation

Components

Thermocouple Datalogger
Stafford & Perez-Alvarez
A datalogger for acquiring temperature measurements using thermocouples and
in real-time.

Automated Relay Switch
Stafford & Perez-Alvarez
An automated relay switch for programmable AC control.

Liquid Exfoliation Cooling
Stafford & Perez-Alvarez
A low-cost, 3D-printed cooling system for high shear exfoliation.

Thermocouple Datalogger

INSTRUCTIONS.md

Build Instructions and Notes

Operating Principles

Thermocouples measure temperature at a junction between two dissimilar metals/conductors using the Seebeck effect. In thermocouple acquisition systems, there are two main requirements. Firstly, the voltage generated by the hot junction temperature of the thermocouple is low and requires amplification. Secondly, this voltage does not provide a reading of temperature by itself. Absolute temperature measurements at the hot junction are obtained using a reference sensor that measures a known temperature. This is known as cold junction compensation. In this build, we use the MAX31855 amplifier which includes cold junction compensation. This is designed for Type-K thermocouples and -200°C to +1350°C output in 0.25°C increments. We connect multiple amplifiers to a prototype HAT (Hardware Attached on Top) that can mount directly onto the header of a Raspberry Pi single board computer. This is then used to communicate with the hardware through the GPIO, record the measurements, and display the temperatures in real-time on a computer screen.

Bill of Materials

|Name |QTY|Description | |:----
|Adafruit Perma-Proto HAT witho
thermocouple amplifier board fo
|Thermocouple cable|1 |Length
Wiring
Soldering Iron

Wiring

Thermocouple amplifier to

A schematic of the connections b
been designed to operate using s

Thermocouple-Datalogger / INSTRUCTIONS.md

Preview | Code | Blame | 170 lines (126 loc) · 8.33 KB | Code 55% faster with GitHub Copilot

record the measurements, and display the temperatures in real-time on a computer screen.

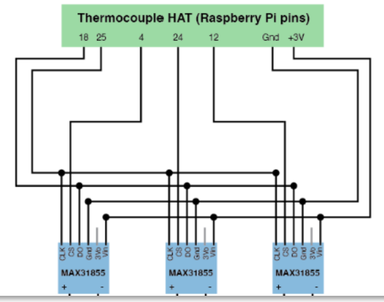
Bill of Materials

Name	QTY	Description
Raspberry Pi	1	Raspberry Pi 3B+ single board computer.
PCB HAT	1	Adafruit Perma-Proto HAT without EEPROM.
Raspberry Pi Case	1	Case for RPi and thermocouple HAT.
Thermocouple Amplifier	3	MAX31855 thermocouple amplifier board for Type-K thermocouples.
Thermocouple connectors	3	For quick connection of thermocouples to datalogger.
Thermocouple cable	1	Length depending on requirements.
Thermocouples	3	Type-K thermocouples to suit the application.
Misc.		• Wiring • Soldering Iron and solder • Small screwdriver

Wiring

Thermocouple amplifier to PCB HAT

A schematic of the connections between the MAX31855 amplifier pins (3) and the Raspberry PI GPIO pins is shown below. The thermocouple HAT has been designed to operate using software SPI.



GitHub repository

Brown, J., Stafford, J., & Perez-Alvarez, D. (2023). OpenLPE. Open Science Framework. https://doi.org/10.17605/OSF.IO/EVUPY

DOCUMENTATION RESOURCES

- READMEs
  - Guide to writing “readme” style metadata
  - README template
  - Documentation methods University of Bath
- Data Dictionaries
  - How to Make a Data Dictionary
  - Data Dictionary Template
- Codebooks
  - Codebook Cookbook
  - Sample Questionnaire with Coding
- Commented Code
  - Coding and Comment Style

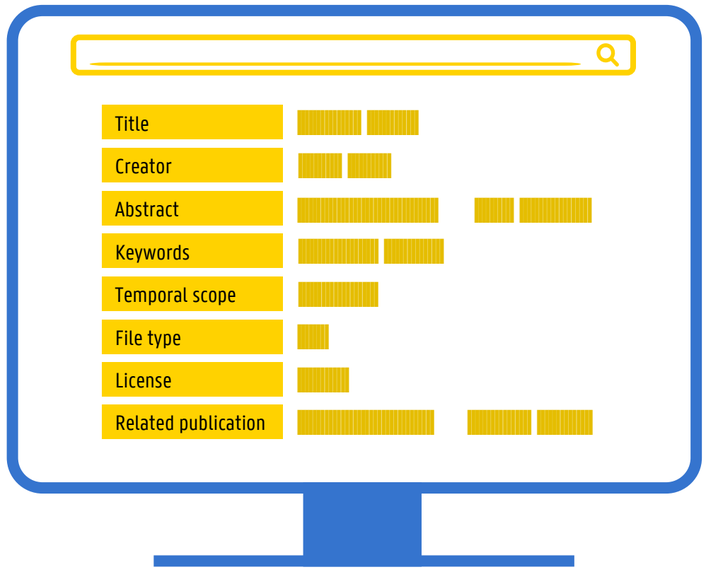
https://www.ugent.be/en/research/datamanagement/during-research/documentation.htm

DOCUMENTATION VS. METADATA

DOCUMENTATION: any contextual information that allows data to be understandable and reusable.

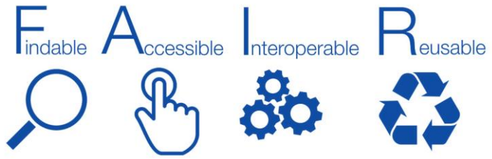


METADATA: information about data, recorded in a **highly structured** way, using a set of defined information fields or elements to be **machine readable (standardized)**.





- Metadata allows others:
  - To find data
  - To assess whether it is useful
  - To know whether they can reuse it, and under which conditions (license), and how it can be accessed
- Documentation allows others:
  - To understand where the data came from and how to (re)use it





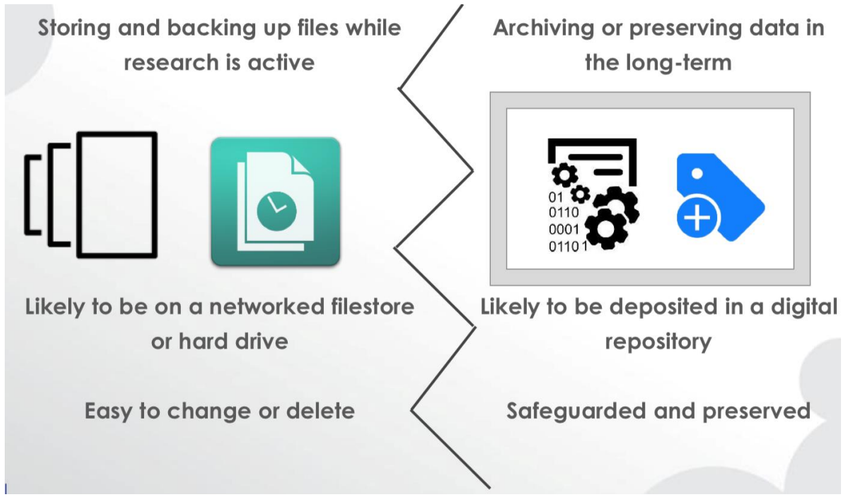
Preserving and sharing

STORING VS. PRESERVING

Keep in mind the difference between storing & preserving data!

- Storing:
  - Active data, while you are still busy with the project.
  - Easy and quick access needed.

- Preserving:
  - At rest, non-active data.
  - Preserving ≠ not deleting data
  - Requires preparation & curation:
    - Ensure long-term accessibility, keep data understandable and reusable



From '[What you need to know to prepare a data management plan]' by Marjan Grootveld, licensed under [CC-BY]

WHERE TO PRESERVE DATA?

- Suitable location -> main purpose of preservation? (verification, internal or external reuse).

Data repository

- Recommended / preferred practice.
- *The* option when goal is sharing & reuse.

In house

- Sometimes it is not be possible/appropriate to deposit data in an external repository, e.g. for legal, ethical, contractual, practical, or other reasons.
- Data needs to be preserved for verification purposes
- Best in-house option: project shares.

Physical samples

- E.g. Molecule archive

WHAT IS DATA SHARING?



- Sharing data from
  - → completed (parts of) research projects
  - → outside your project or research team.
- Not
  - exchanging data with collaborators while your research is active.
  - Share data with your promotor

- Sharing data can have many **advantages** for your career, for science and society at large.

WAYS OF SHARING DATA

Depositing in a repository explicitly required by some funders and some journal policies

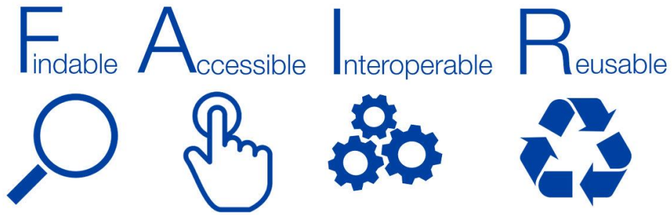
Upon request

Personal or project website

Article suppl. material

Data repository or archive

Research shows that almost non of the data that is requested is actually shared (<11%[1], <7%[2])



1 Federer, Lisa M et al. (2018) https://doi.org/10.1371/journal.pone.0194768

2 Gabelica et al. (2022), https://doi.org/10.1016/j.jclinepi.2022.05.019

WHAT IS A DATA REPOSITORY?

A database infrastructure that collects, manages and stores data for preservation and sharing

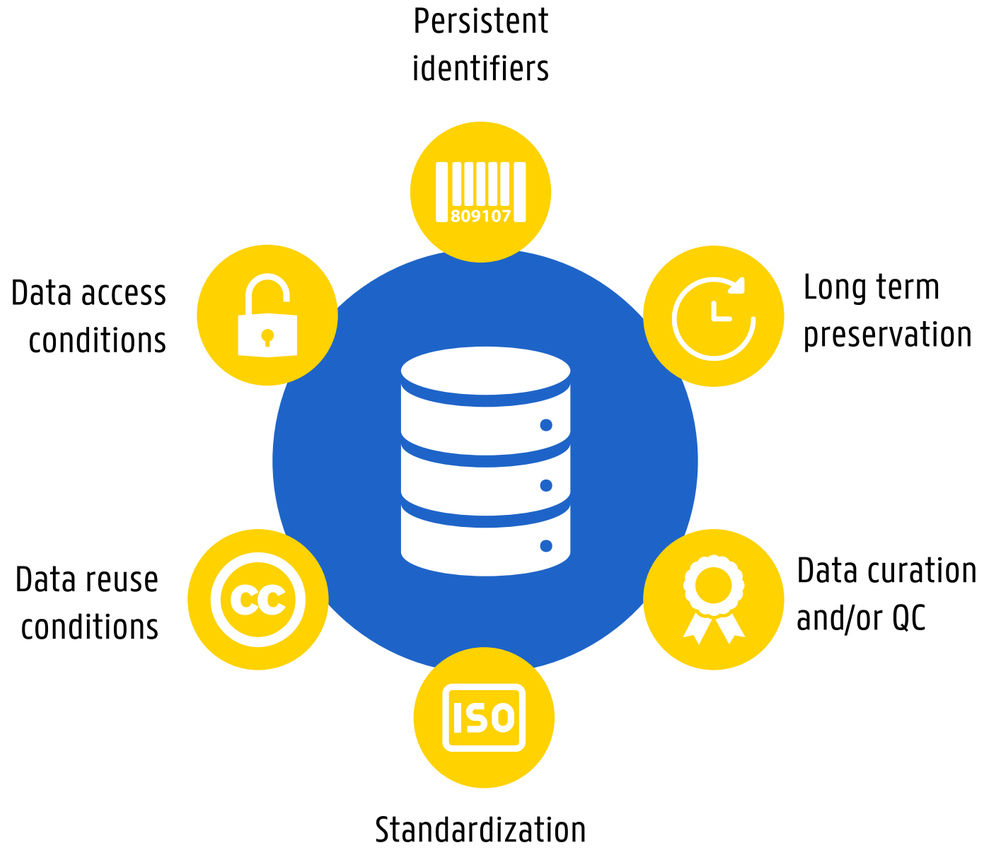
I want to share my data

Researcher

Data

Metadata

Documentation



I want to find and re-use data

Scientific community

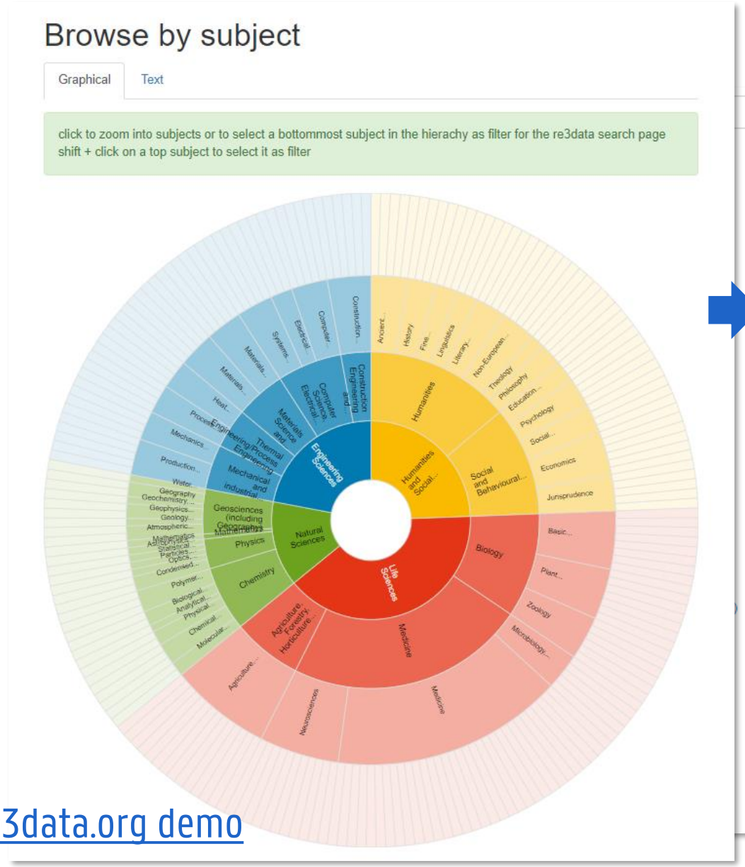
Search & filter

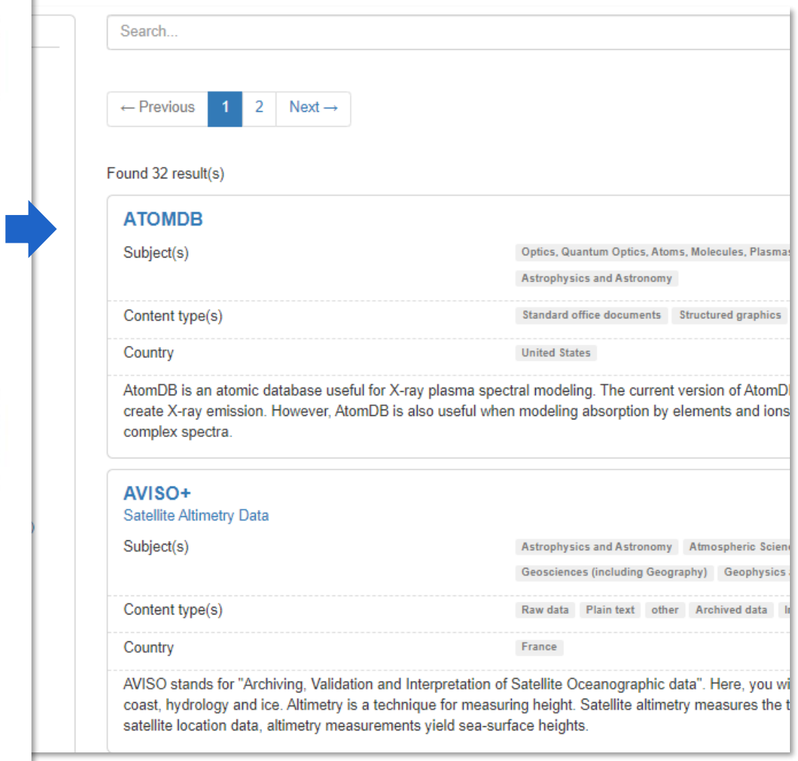
View

Download / Export

HOW TO SELECT A DATA REPOSITORY?

1. Where do your peers publish data? Domain's consensus -> domain repositories
2. Does the journal / funder recommend or require certain repositories?
3. Registries of data repositories: re3data.org, FAIRsharing
4. No domain repository? -> General purpose repositories (Zenodo, Dryad, B2SHARE…)





re3data.org demo

Data Repository Guidance

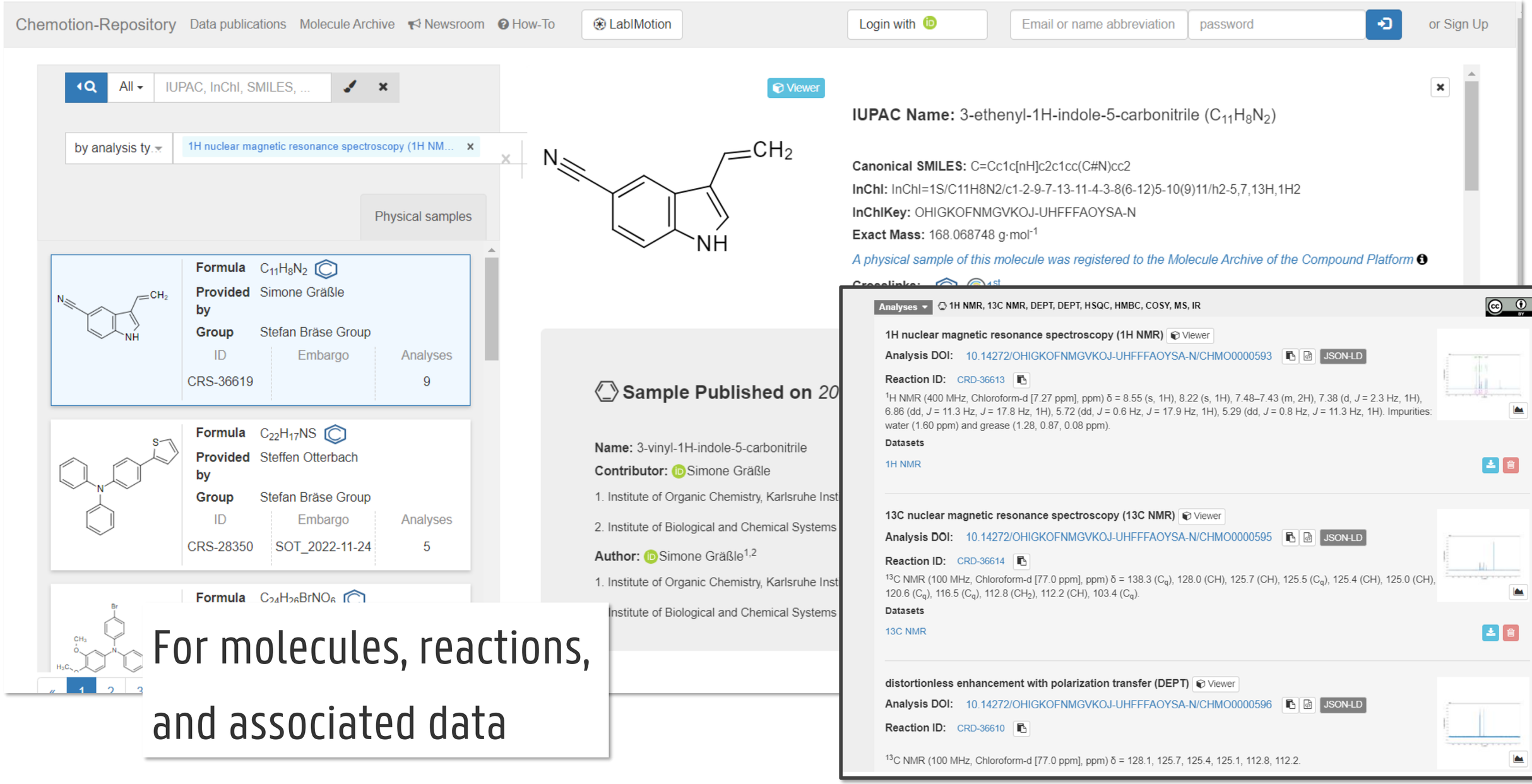
Scientific Data mandates the release of datasets accompanying our Data Descriptors.

View data repositories

- **Biological sciences:** Nucleic acid sequence; Protein sequence; Molecular & supramolecular structure; Neuroscience; Omics; Taxonomy & species diversity; Mathematical & modelling resources; Cytometry and Immunology; Imaging; Organism-focused resources
- **Health sciences**
- **Chemistry and Chemical biology**
- **Earth, Environmental and Space sciences:** Broad scope Earth & environmental sciences; Astronomy & planetary sciences; Biogeochemistry and Geochemistry; Climate sciences; Ecology; Geomagnetism & Palaeomagnetism; Ocean sciences; Solid Earth sciences
- **Physics**
- **Materials science**
- **Social sciences**
- **Generalist repositories**

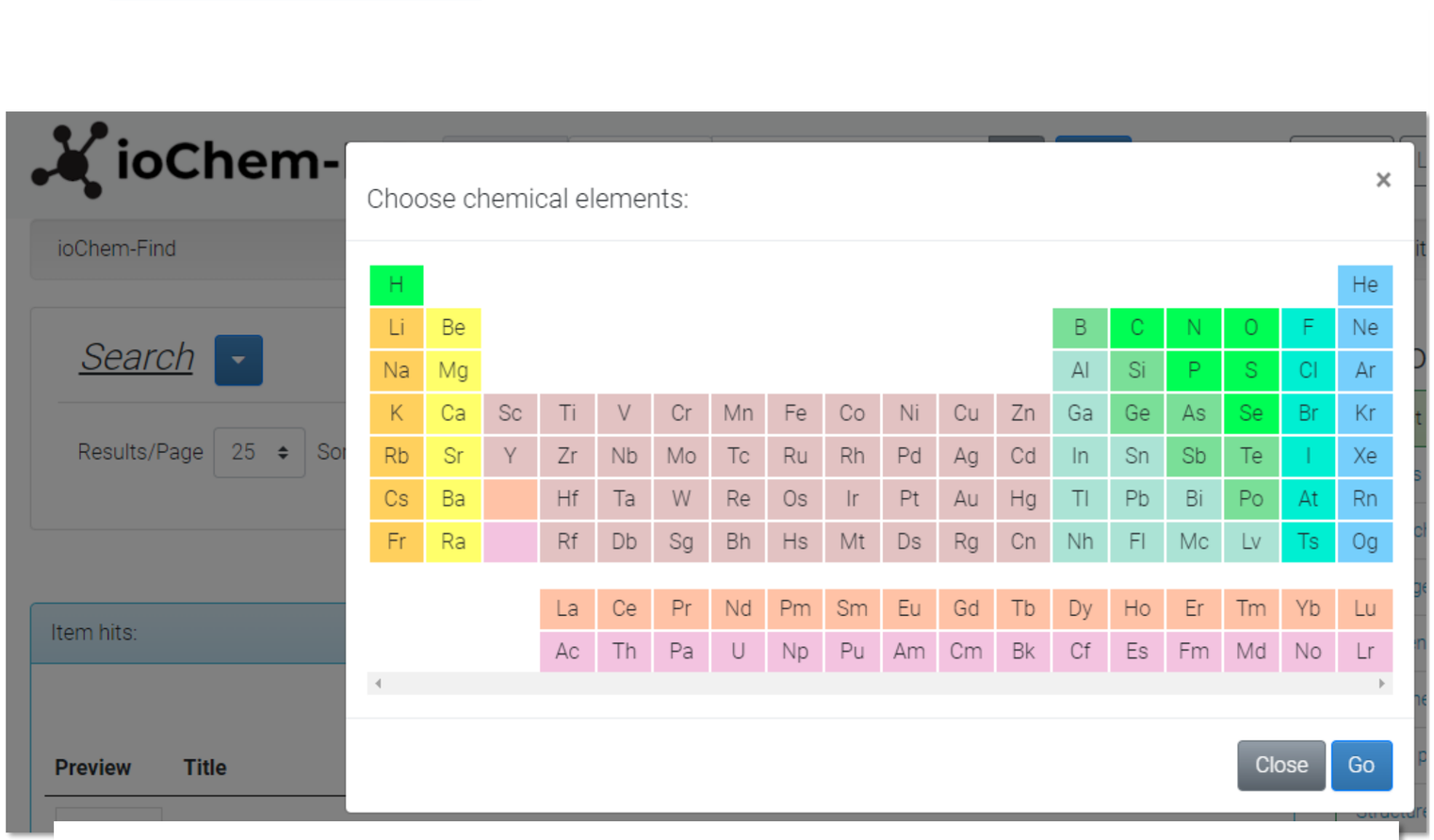
Data repositories recommended by Nature

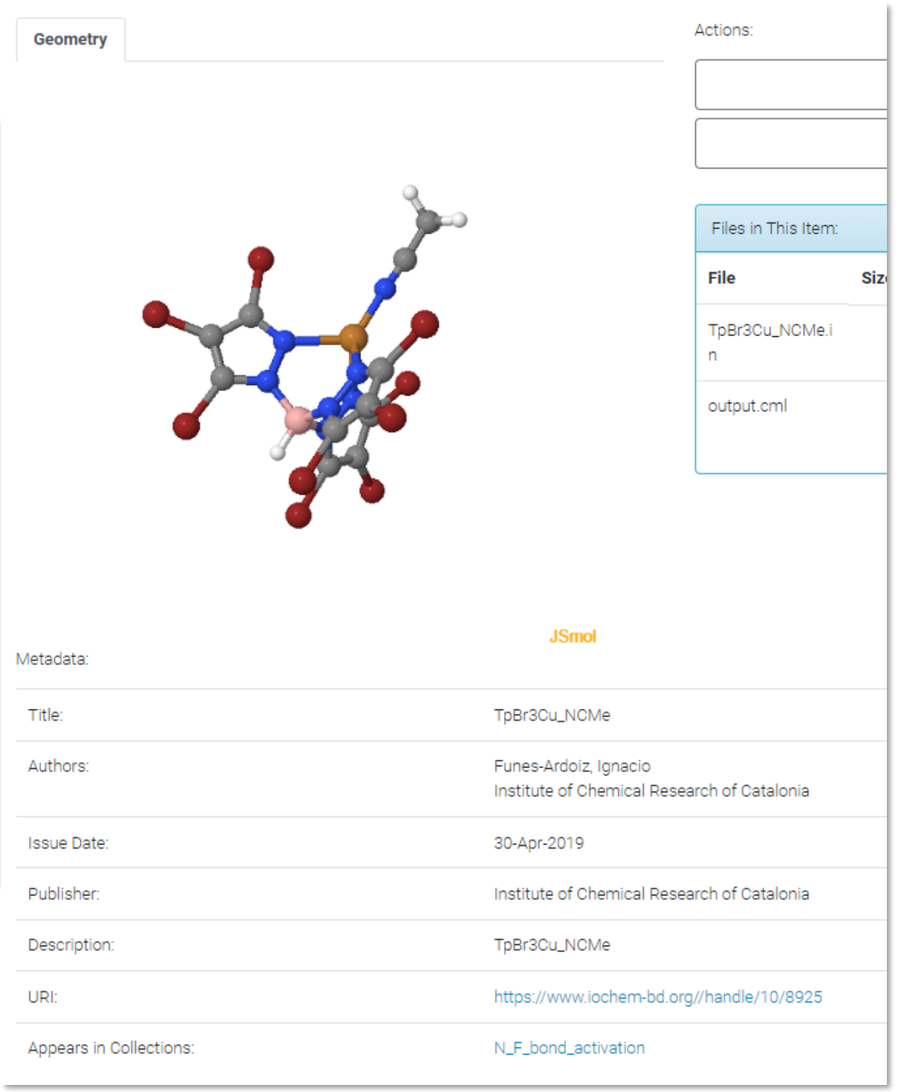
CHEMOTION



For molecules, reactions, and associated data

IOCHEM-BD



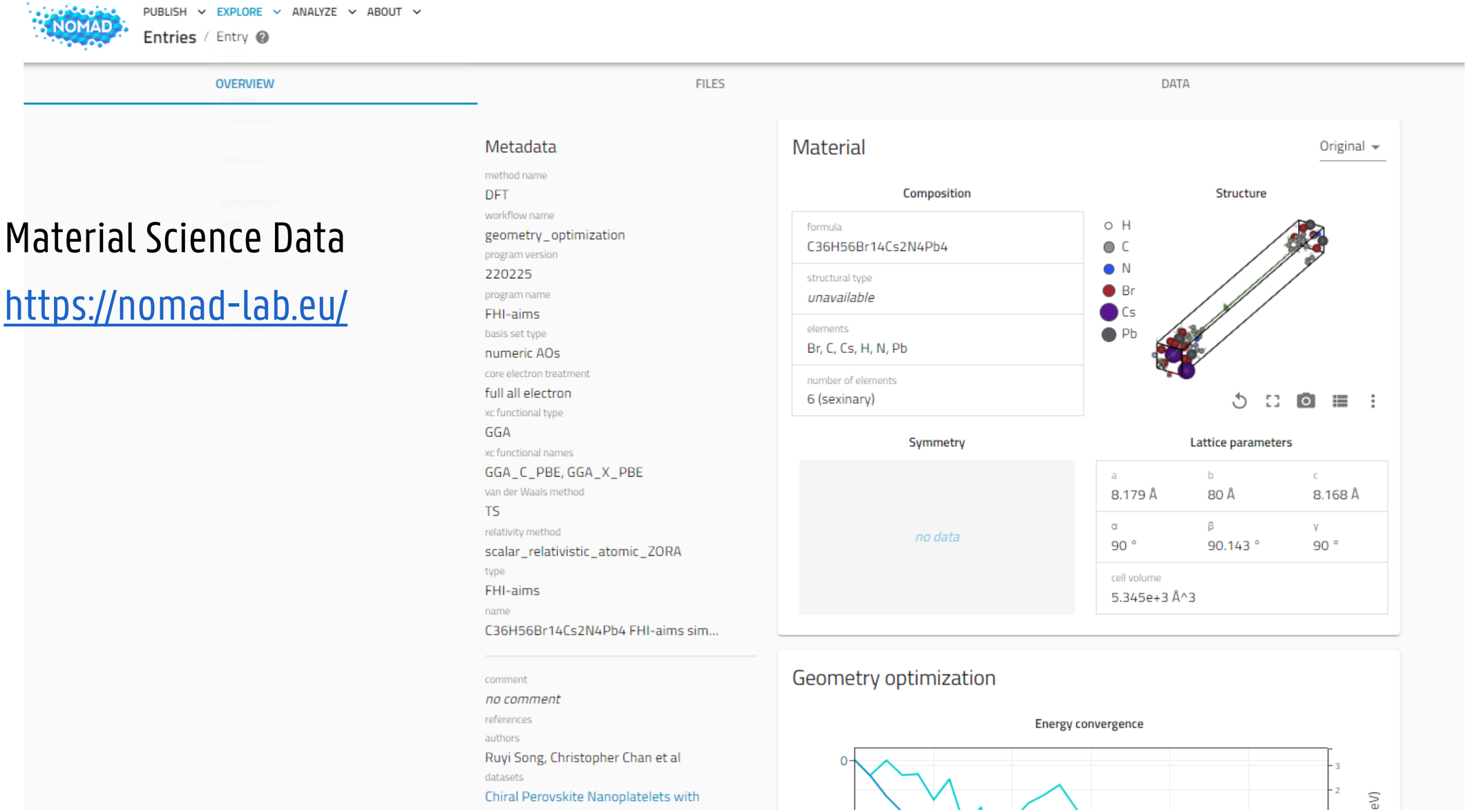


Computational chemistry

NOMAD

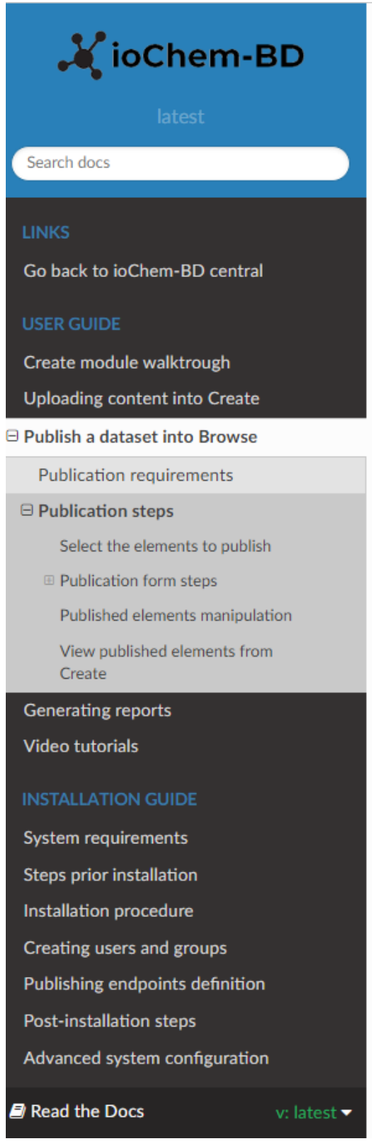
Material Science Data

https://nomad-lab.eu/



CHECK REPOSITORY SUBMISSION GUIDELINES

Be prepared to meet possible requirements and save time at the end of the project: formats, documentation & metadata.



Once published, both elements are linkable, searchable and viewable publicly.

Publication requirements

To perform calculation publication, users must have defined (on Browse module) a **community where to publish**. The steps required to create a community and assign its administrator falls upon ioChem-BD's system administator and are described on publishing endpoints definition page.

Publication steps

Select the elements to publish

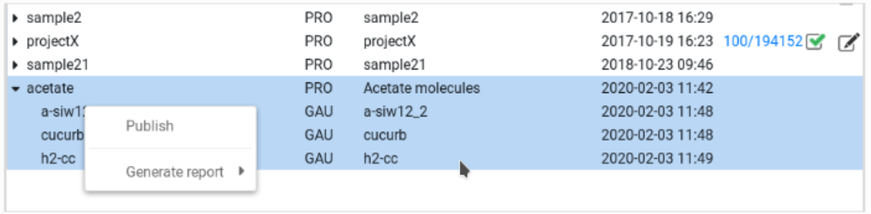
On Create module, navigate to the project where the desired calculations reside.

We can select an individual calculation by clicking on it, or using multiselect with *Ctrl + Click*, *Alt + Click*. Another way to select calculations is right clicking on parent project and choose *Select child elements* from its context menu.

Attention

You must select the project **and its inner calculations** prior publishing.
If you only select the project and not its content you will publish the project **empty of contents**.

Once desired projects and calculations are selected, right click on the current selection to display its context menu. Then select *Publish* option.



Calculation selection

If an error message appears at this point, please read the FAQ entry about no publishing community defined.

https://docs.iochem-bd.org/en/latest/index.html

DATA REPOSITORIES & FAIR DATA

→ Key infrastructures that provide functionalities to make your data FAIR.

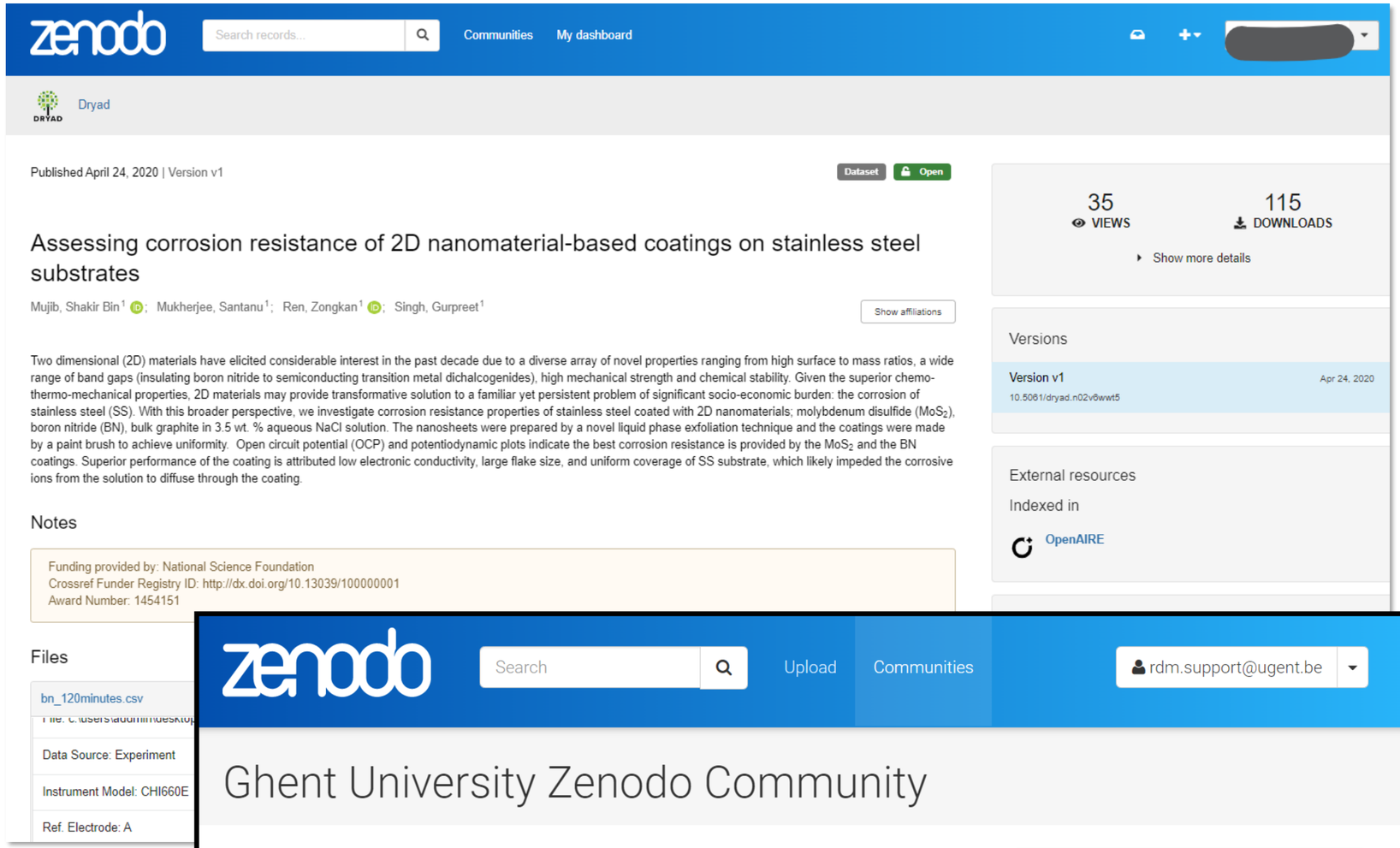
→ Shared responsibilities between repositories and researchers in making data FAIR

FAIR principle	In practice	Repository	Researcher
FINDABLE	Assign a persistent identifier (PID)	●	
	Data are described with rich metadata	●	●
	Metadata include the PID, are indexed & searchable online	●	
ACCESSIBLE	(Meta)data retrievable via PID using a standardised communications protocol (e.g. html)	●	
	Authentication and authorisation procedure	●	
INTEROPERABLE	Metadata standards	●	
	Controlled vocabularies	●	●
	References to other (meta)data	●	●
REUSABLE	(Machine readable) license / terms of use	●	●
	Detailed provenance documentation		●
	Domain standards	●	●

Shared responsibilities between repositories and researchers in making data FAIR

GENERAL PURPOSE REPOSITORIES & UGENT ZENODO COMMUNITY

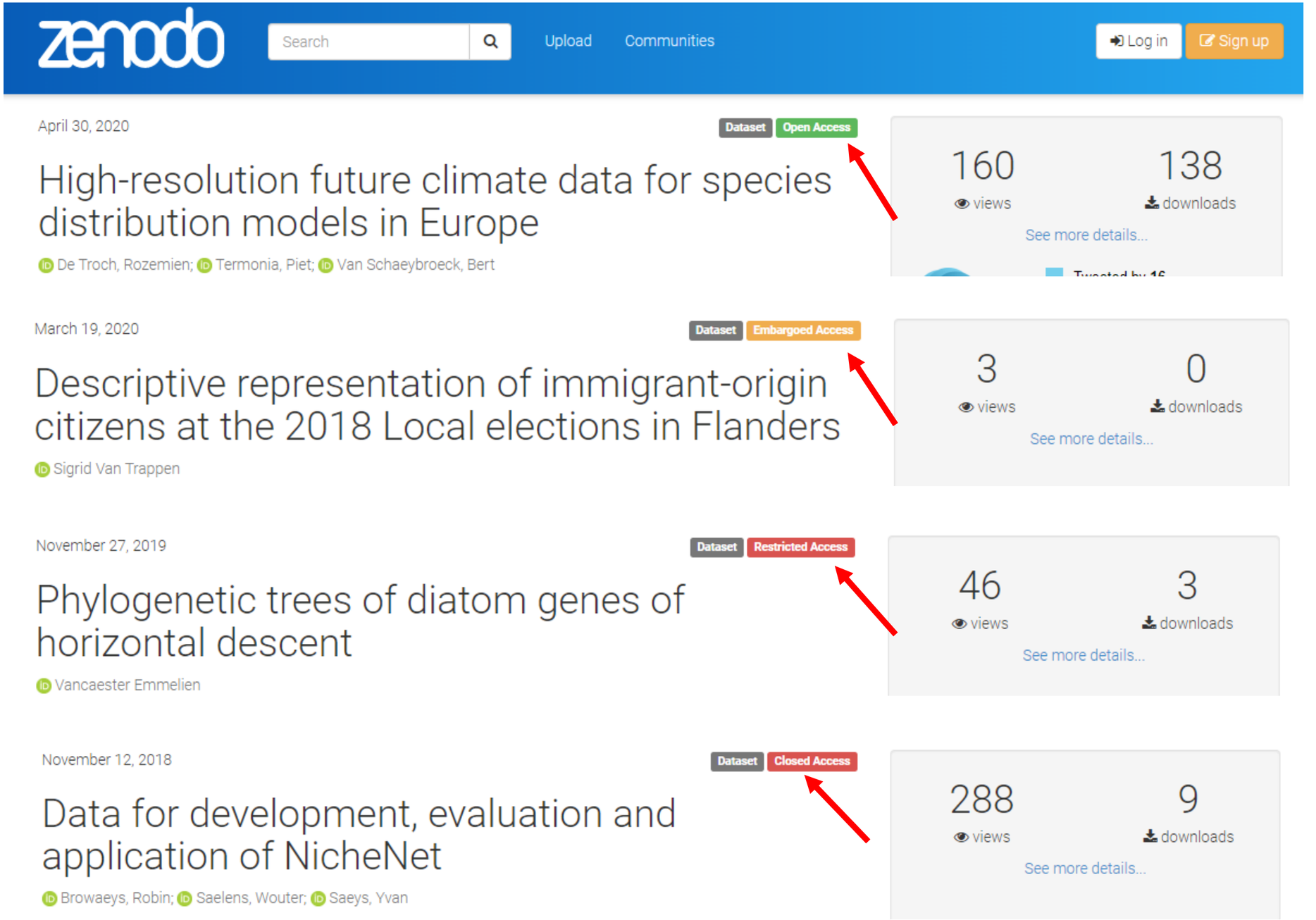
- Base FAIRness level.
- Ugent Zenodo Community
  - Accepts research data and software
  - Ugent data curators
  - Better ranking results
  - Avoid dataset size restrictions
  - Avoid spam filters



ACCESSIBLE ≠ OPEN

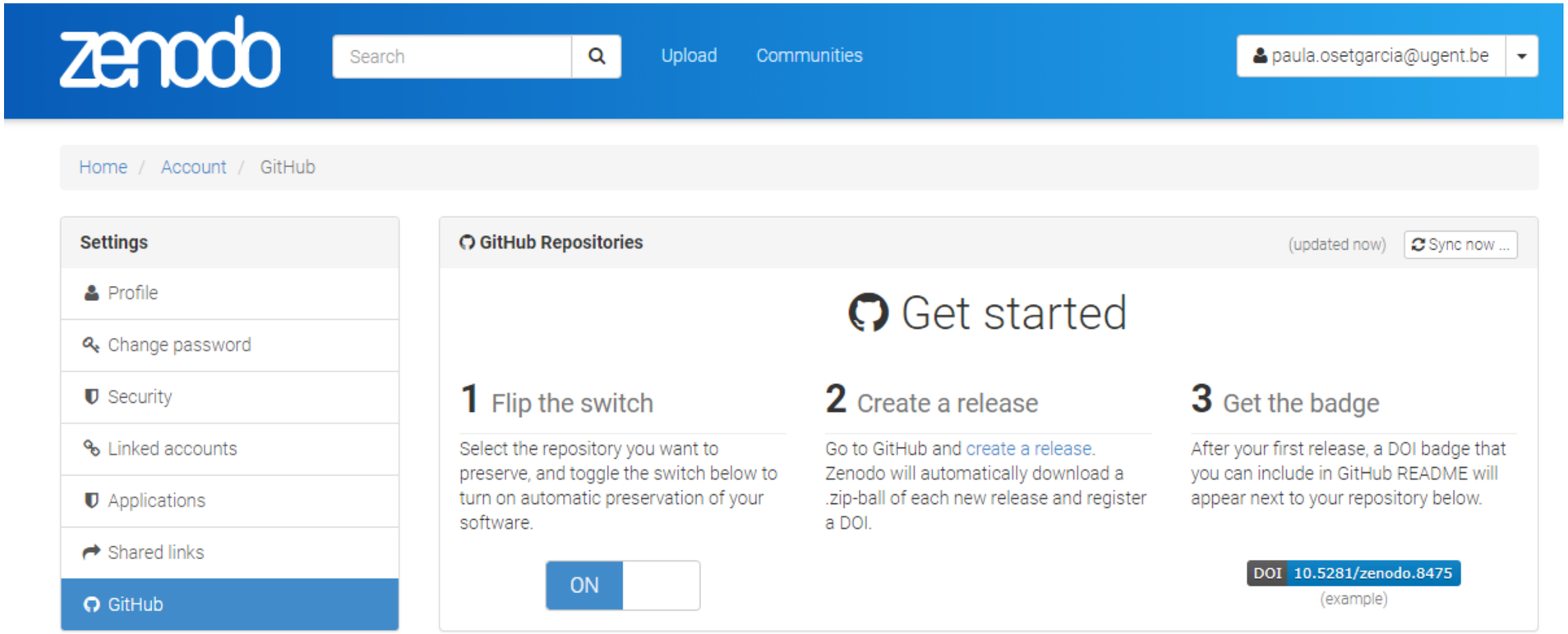
There are legitimate reasons why some data cannot be (fully open):

- Personal data (GDPR)
- Valorization
- Other types of confidential / sensitive data
- Third party data
- Contracts/agreements



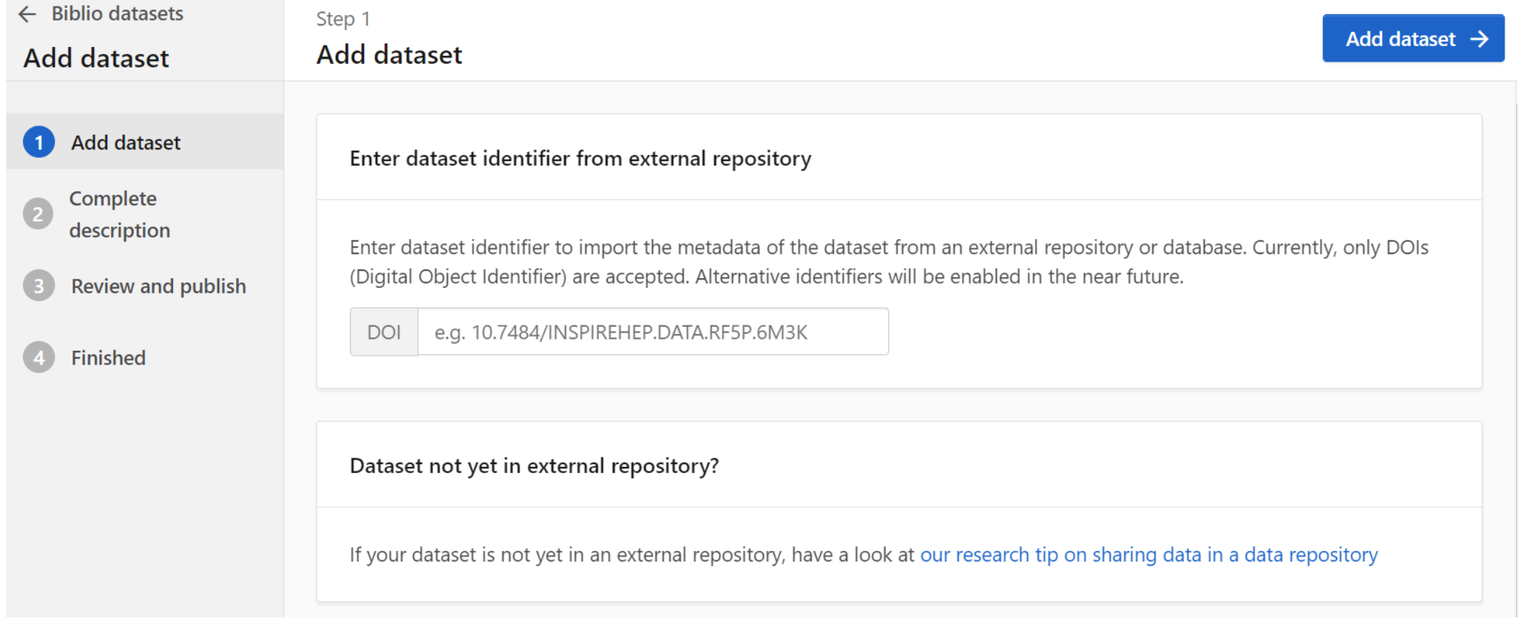
OTHER RESEARCH OUTPUTS
SHARING CODE AND MAKING CODE CITABLE

- GitHub – Zenodo integration: https://guides.github.com/activities/citable-code/



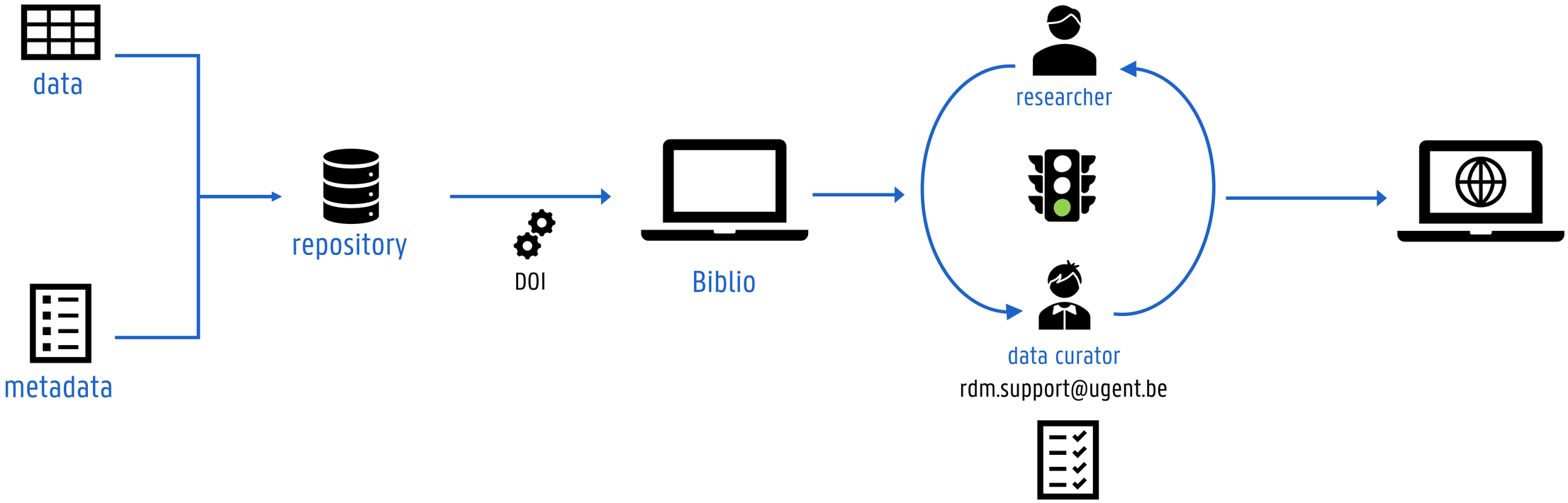
BIBLIO: DATASETS AND SOFTWARE

- Ugent Scholarly Publication policy (2023)
- Register datasets in Biblio using DOI
- Automatically harvests several metadata fields
- Enhance findability of your research outputs



Research tip Biblio: how to register my datasets?:

BIBLIO REGISTRATION WORKFLOW FOR DATASETS AND SOFTWARE



DATA CITATION

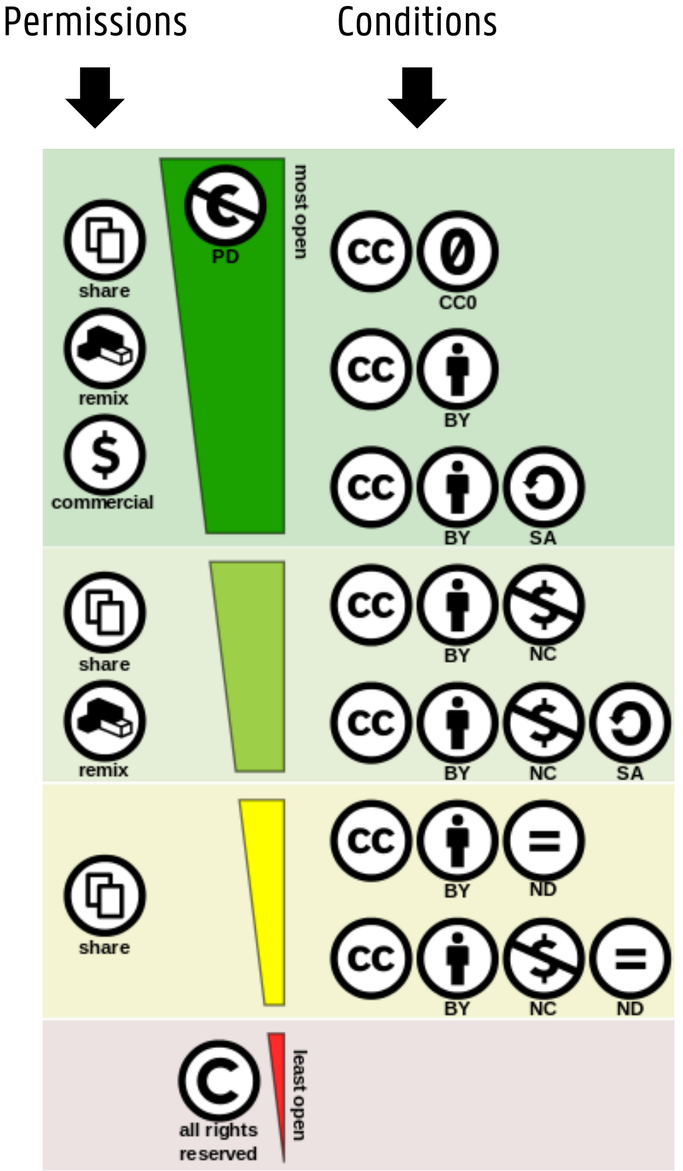
- Providing a reference to data/software in the same way as researchers routinely provide a bibliographic reference to other scholarly resources.
  - Whenever a claim relies upon data, the corresponding data should be cited.
- **"Data as first class research output".**
- Plays a similar role as a bibliographic citation:
  - To assign due credit: citation contains name(s) of responsible agent.
  - To track impact: it is accompanied by unique identifier.
  - To locate and access resource: citation contains sufficient information.
- Give attribution even if not a legal requirement (e.g. CC-0).

Properly cited, not as a footnote!

LICENSES

- Licenses: means for owners of rights (in datasets) to **grant permission(s)** under certain conditions.
- Define what can and cannot be done with data in a clear way.
  - Nature: expand rather than restrict what can be done.

    E.g, data may be 'publicly available' but, without license, users do not have permission to access, use and share it under copyright or database laws.
- Licenses are **not automatic.**
- Avoid custom-made licenses / agreements as much as possible.
- Standard licenses: terms and conditions are already defined.
  - Creative Commons, Open Data Commons, open software licenses



Creative commons license spectrum, CC BY 4.0

THE BIG PICTURE

Plan for data management, keep your DMP updated

Explore potential data repositories and their requirements & guidelines.

- Inc. use of standards, vocabularies, etc.

re3data.org
REGISTRY OF RESEARCH DATA REPOSITORIES

FAIRsharing

DCC

Adopt domain standards during data collection & processing

Document as you go

- Notebook
- Codebook, data dictionary
- Workflows
- Protocols
- Readme files
- Data overview with metadata
- File organization and naming

Deposit data & documentation in repository; describe it with rich metadata, inc. access & reuse conditions

Register data in Biblio

TAKE-HOME MESSAGES

- Managing data: good for you, for research, for society
- Start with planning: DMPs as a segway to good management practices
- Have reuse in mind (think as a re-user) and implement good practices through the different steps of the research lifecycle